Relay-Embedded Fault Classification in Power Distribution Networks Using Deep Learning Methods

Project Timeline

Weeks	Phase / Task	Description	E/21/143	E/21/145	E/21/327
1 – 3	**Literature Review & Tool Familiarisation**	Survey on Fault Classification. Survey ML/DL advances for waveform feature extraction. Explore simulation environments and verify time-series data export pipelines.	Refer Fault Classification papers published before. Set up simulation interface. Test tool compatibility and environment configuration.	Refer Fault Classification papers published before. Review ML/DL architectures suitable for fault classification.	Refer Fault Classification papers published before. Investigate data export methods. Confirm time-series extraction reliability.
4 – 6	**Independent Network Modelling**	Build three separate, fully functional test feeder network models. Each includes distributed generation sources (synchronous and inverter-based).	Build Network Model 1.	Build Network Model 2.	Build Network Model 3.
6 – 7	**Fault Injection & Dataset Generation**	Inject thousands of faults (3-phase, line-to-line, line-to-earth) at varied positions across all three models. Extract raw voltage and current waveform time-series.	Develop and run automation scripts for systematic fault injection.	Define and control extraction parameters to ensure data consistency.	Aggregate and organise the full combined dataset from all three models.
Mid-Semester					
9 – 10	**Data Preprocessing**	Clean and structure the datasets. Apply mathematical transforms to create training/testing splits.	Format and structure data for compatibility with ML pipelines.	Lead advanced preprocessing. Apply signal transforms (e.g. FFT, DWT).	Implement conventional relay logic models as baseline benchmarks.
	Conventional Logic	Implement traditional relay protection algorithms for baseline comparison.			
11 – 12	**ML/DL Model Training & Testing**	Feed preprocessed fault data into the selected ML/DL model. Train, validate, and refine performance across thousands of fault scenarios.	Evaluate model metrics. Assist with hyperparameter tuning.	Lead model training and validation. Drive architecture decisions.	Evaluate model metrics. Assist with hyperparameter tuning.